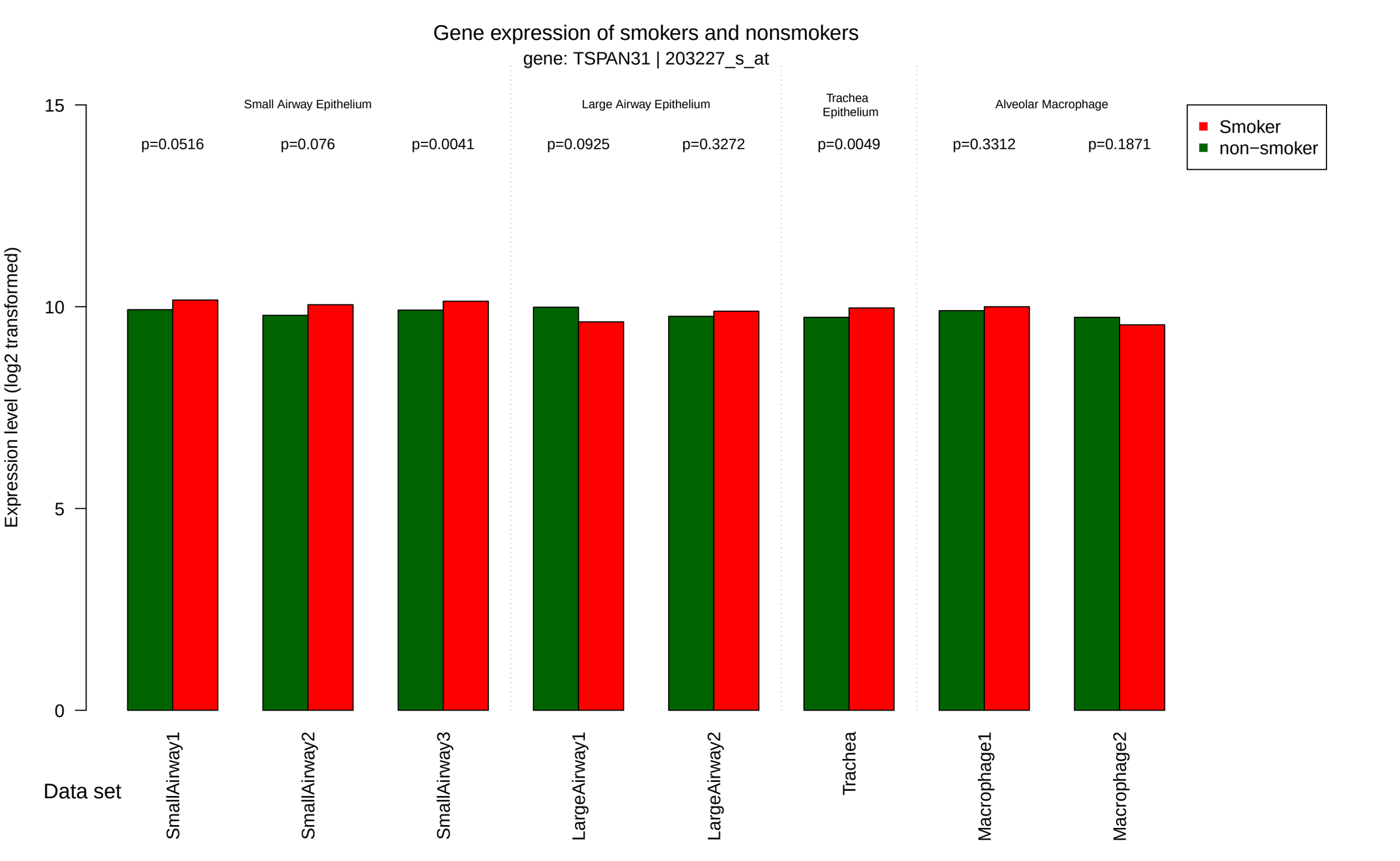
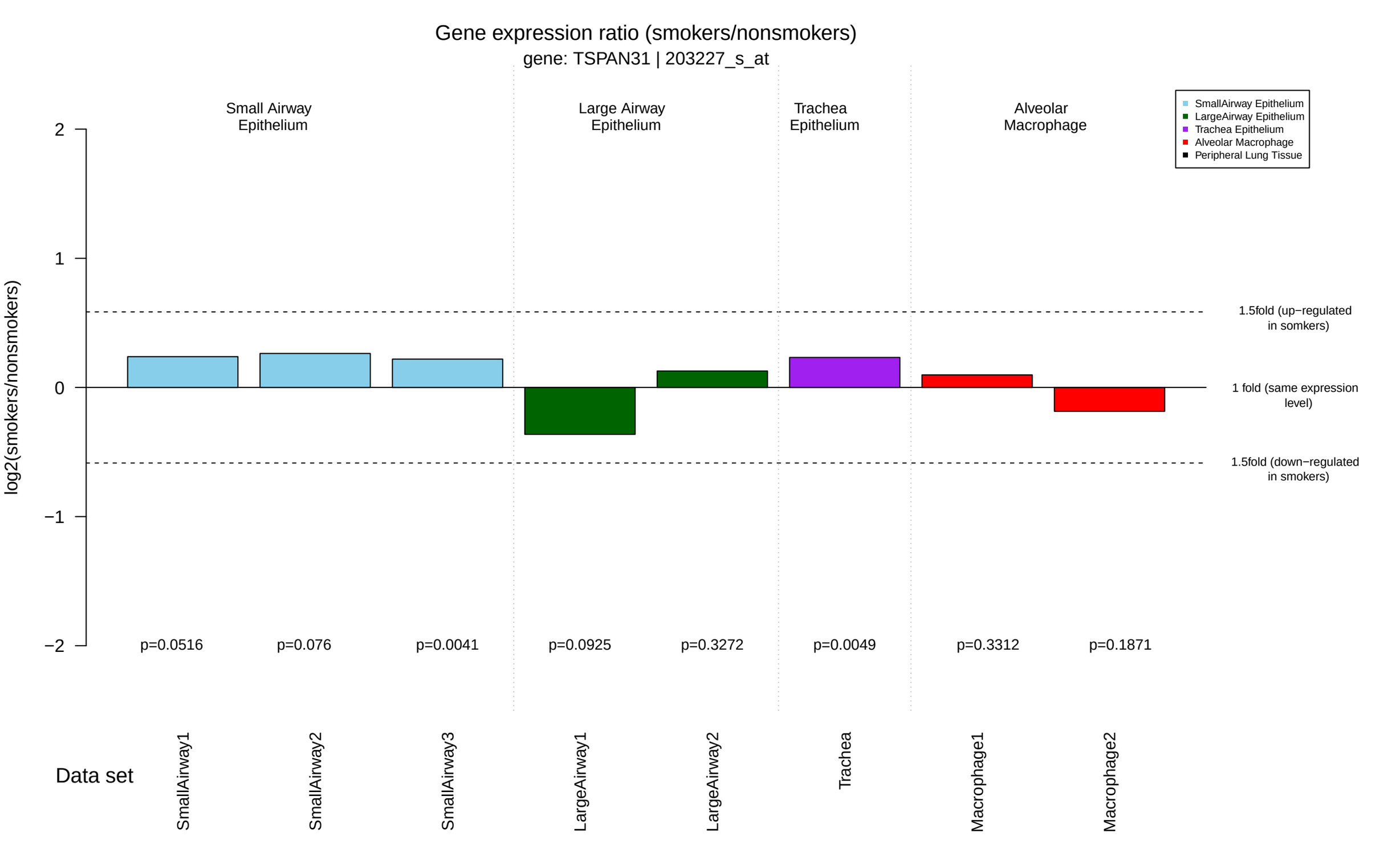
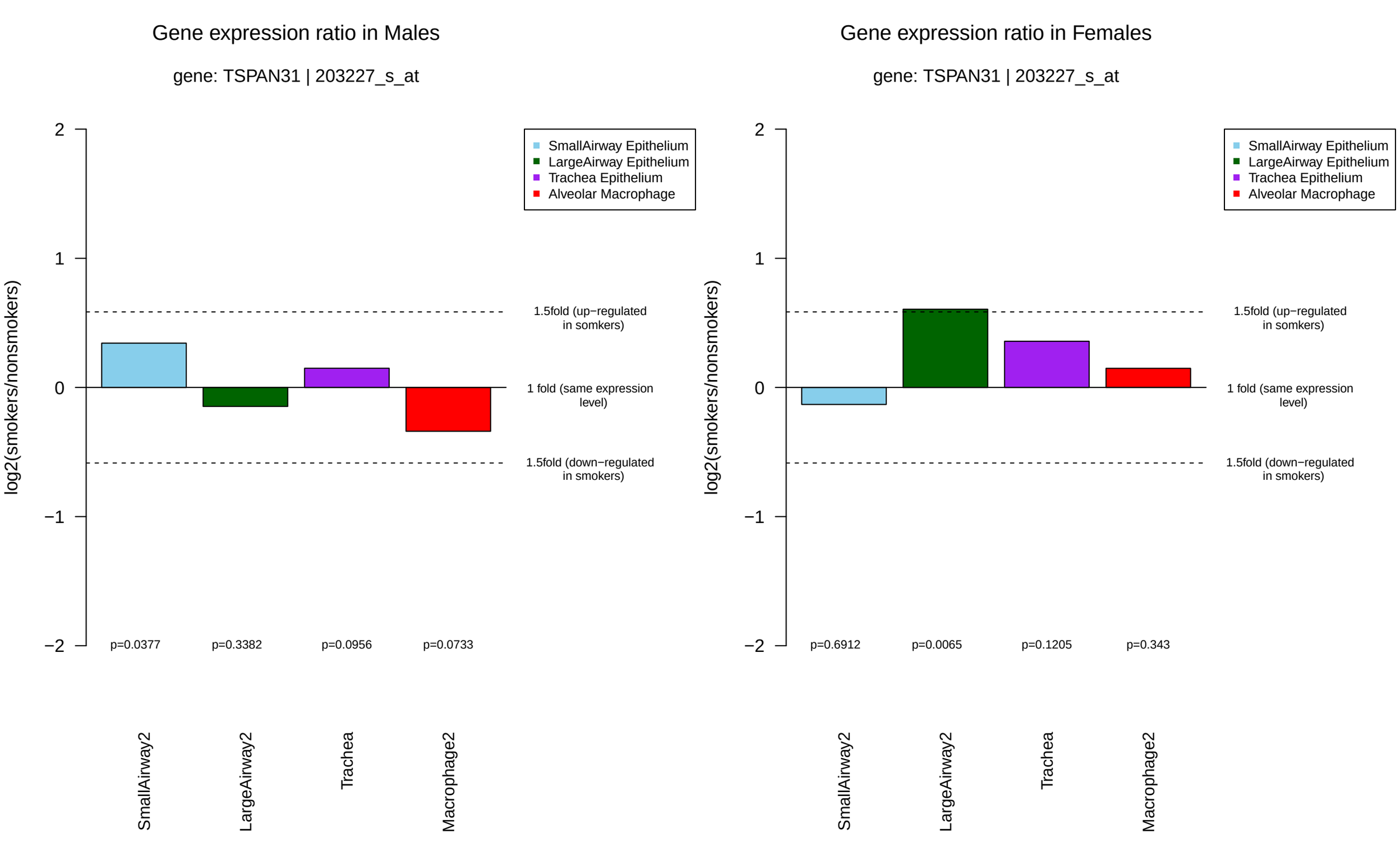
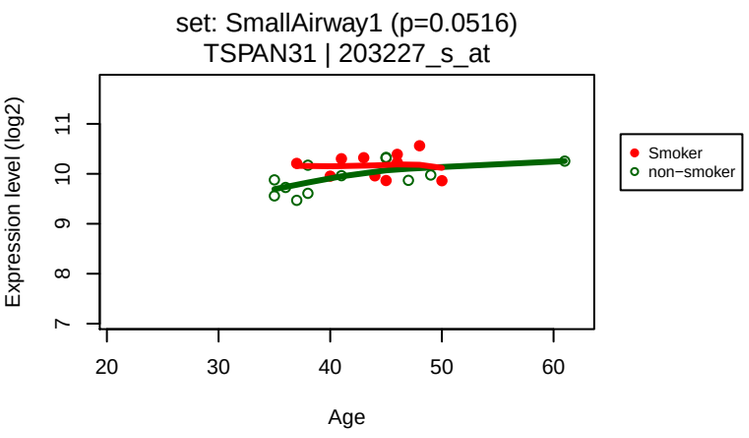
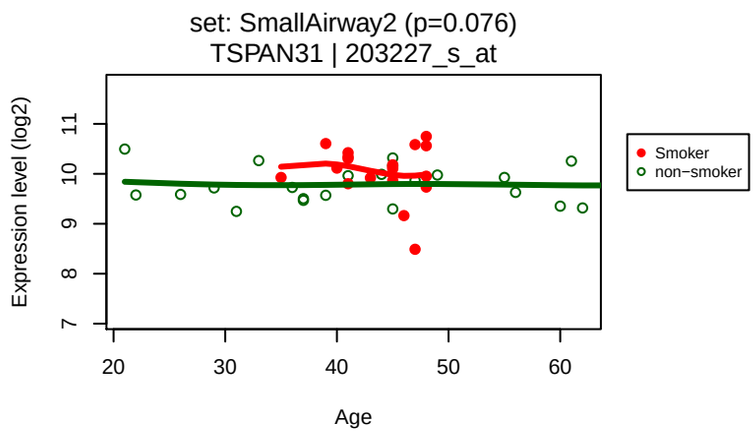
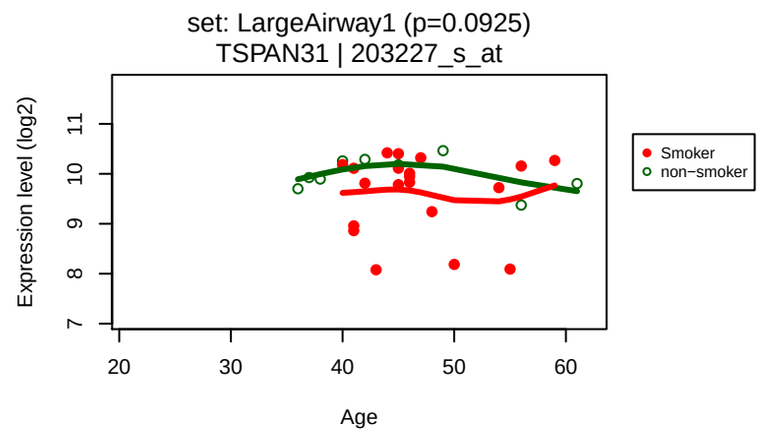

Gene expression of smokers and nonsmokers
gene: TSPAN31 | 203227_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0516
p=0.076
p=0.0041
p=0.0925
p=0.3272
p=0.0049
p=0.3312
p=0.1871
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

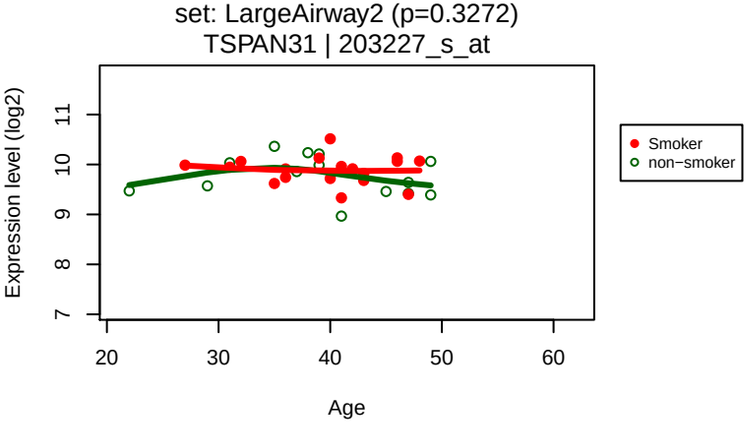

Gene expression ratio (smokers/nonsmokers)
gene: TSPAN31 | 203227_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0516
p=0.076
p=0.0041
p=0.0925
p=0.3272
p=0.0049
p=0.3312
p=0.1871
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

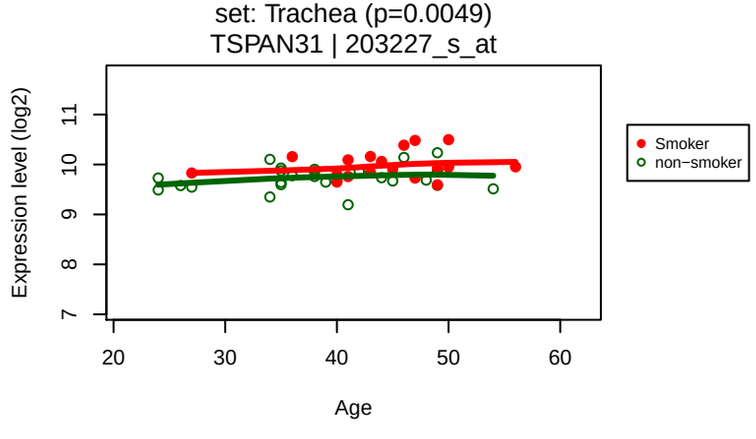

Gene expression ratio in Males
gene: TSPAN31 | 203227_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0377
p=0.3382
p=0.0956
p=0.0733
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TSPAN31 | 203227_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6912
p=0.0065
p=0.1205
p=0.343
SmallAirway2
LargeAirway2
Trachea
Macrophage2

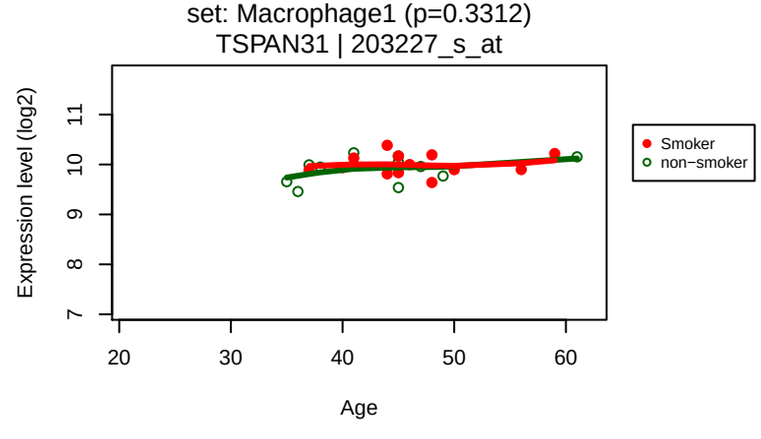

set: SmallAirway1 (p=0.0516)
TSPAN31 | 203227_s_at
Expression level (log2)
Age
Smoker
non−smoker

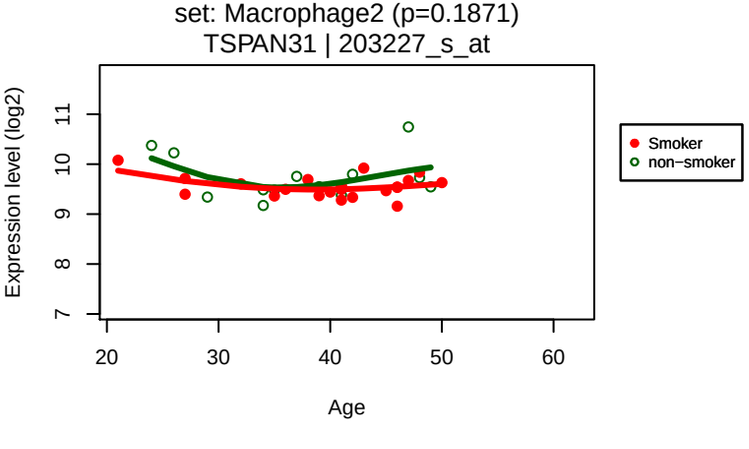

set: SmallAirway2 (p=0.076)
TSPAN31 | 203227_s_at
Expression level (log2)
Age
Smoker
non−smoker

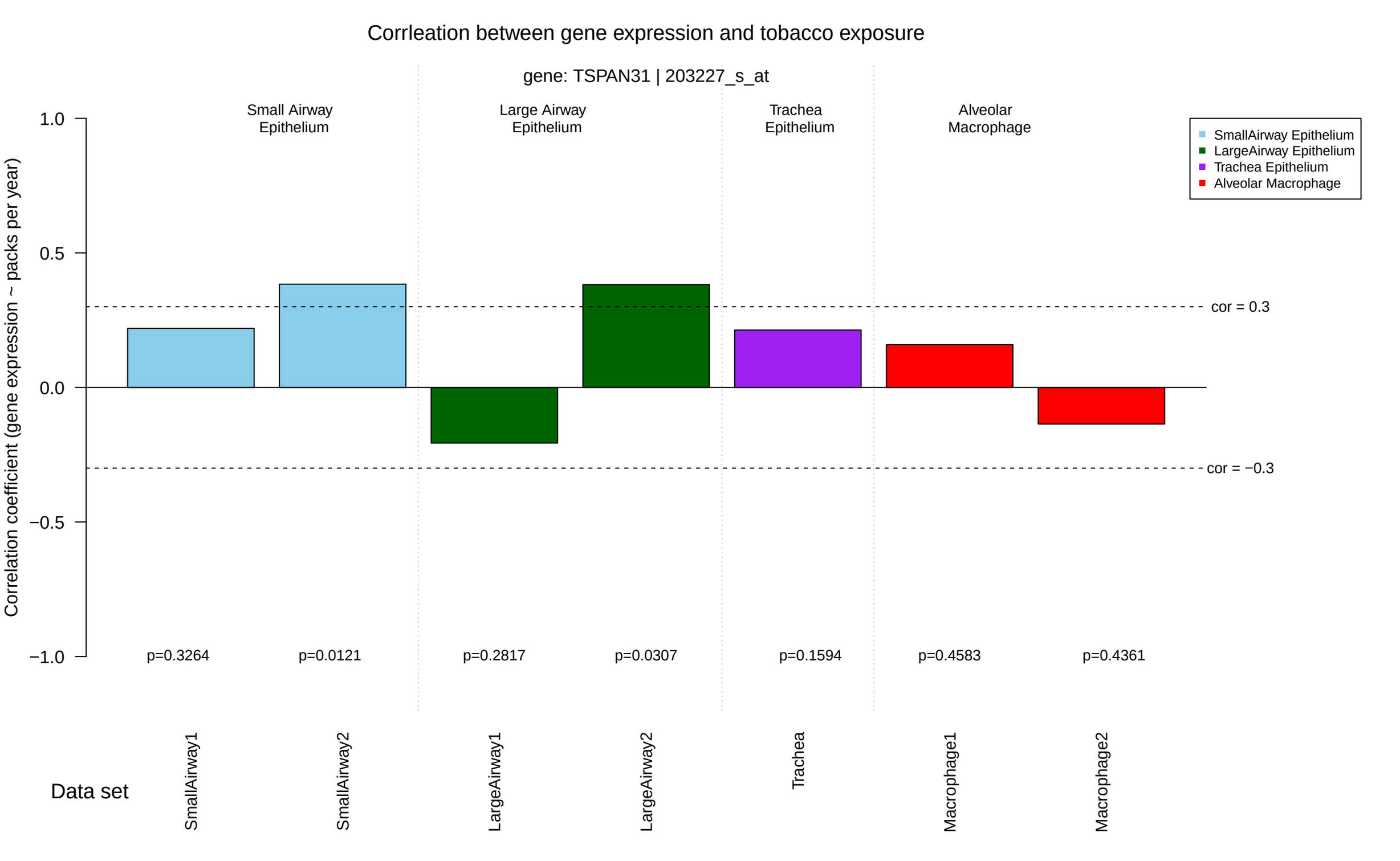

set: LargeAirway1 (p=0.0925)
TSPAN31 | 203227_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3272)
TSPAN31 | 203227_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0049)
TSPAN31 | 203227_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3312)
TSPAN31 | 203227_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1871)
TSPAN31 | 203227_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TSPAN31 | 203227_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3264
p=0.0121
p=0.2817
p=0.0307
p=0.1594
p=0.4583
p=0.4361
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2